

Fluid-Damp Gear Shift Interface

Cost Analysis per Unit — CSMFAB 035

Category: Aegis Embark — Interior **Unit Basis:** 1 shift knob + MR control unit **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
LORD MRF-140CG fluid reservoir (PEEK housing)	0.15 kg	\$250.00/kg	\$37.50
PEEK CF40 reservoir body	0.2 kg	\$280.00/kg	\$56.00
ZrO2 ceramic top cap insert	0.15 kg	\$250.00/kg	\$37.50
YInMn Blue ZrO2 glaze (top face)	0.02 kg	\$75.00/kg	\$1.50
SmCo electromagnet coil (ceramic-isolated)	0.08 kg	\$120.00/kg	\$9.60
Optical trigger control wire (PMMA fiber)	0.2 m	\$25.00/kg	\$0.50
Universal thread adapter set	0.08 kg	\$45.00/kg	\$3.60
BOM Subtotal			\$146.20

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$146.20	\$119.88	\$95.03
Direct Labor	\$92.00	\$62.00	\$42.00
Overhead (15% of labor)	\$13.80	\$9.30	\$6.30
Quality Control & Testing	\$11.04	\$4.96	\$2.10
Packaging & Logistics	\$25.00	\$20.00	\$15.00

Tooling & Equipment Notes: Ceramic cap mold: \$4,500 NRE. MRF fill: automated at \$6/unit.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$420	\$672 – \$924	35-55%
Small Batch (100-500/yr)	\$285	\$427 – \$570	33-50%
Production Volume (5K-50K/yr)	\$200	\$280 – \$360	28-45%

Market Reference: \$240-360 (gear knob aftermarket \$30-200; premium MR haptic niche; motorsport + EV segment)

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$26 – \$51 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

End of Fluid-Damp Gear Shift Interface Cost Analysis | CSMFAB 035 | June 2026